

Query Comparison Chart — Native Driver vs Mongoose (ODM)

Operation	Native MongoDB Driver	Mongoose (ODM)
 Connect to DB	<code>MongoClient.connect(uri)</code>	<code>mongoose.connect(uri)</code>
 Create a document	<code>db.collection('users').insertOne({ name: 'Satish' })</code>	<code>User.create({ name: 'Satish' })</code>
 Find all	<code>db.collection('users').find({}).toArray()</code>	<code>User.find()</code>
 Find one	<code>db.collection('users').findOne({ email: 'a@b.com' })</code>	<code>User.findOne({ email: 'a@b.com' })</code>
 Find by ID	<code>db.collection('users').findOne({ _id: ObjectId(id) })</code>	<code>User.findById(id)</code>
 Update one	<code>db.collection('users').updateOne({ _id }, { \$set: { name: 'Ram' } })</code>	<code>User.updateOne({ _id }, { name: 'Ram' })</code>
 Update by ID	<code>db.collection('users').findOneAndUpdate(...)</code>	<code>User.findByIdAndUpdate(id, { name: 'Ram' })</code>
 Delete one	<code>db.collection('users').deleteOne({ _id })</code>	<code>User.deleteOne({ _id })</code>
 Delete by ID	<code>db.collection('users').findOneAndDelete(..)</code>	<code>User.findByIdAndDelete(id)</code>
 Populate/Join	Manual: lookup with aggregation pipeline	<code>User.find().populate('blogs')</code>
 Count documents	<code>db.collection('users').countDocuments({})</code>	<code>User.countDocuments()</code>
 Aggregation	<code>db.collection('orders').aggregate([...])</code>	<code>Order.aggregate([...])</code>
 Validation	Manual, in code	In schema: <code>{ name: { type: String, required: true } }</code>
 Middleware/Hook	 Manual	<code>schema.pre('save', fn)</code>

Operation	Native MongoDB Driver	Mongoose (ODM)
 Schema	 Not enforced	 Strict schema (new mongoose.Schema({...}))
 Model definition	 Use directly: db.collection()	mongoose.model('User', userSchema)

Example Use Case

Native MongoDB Driver

js

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```
const user = await db.collection('users').findOne({ email: 'test@example.com' });
```

Mongoose ODM

js

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```
const user = await User.findOne({ email: 'test@example.com' });
```